

CCNA 200-301 v1.1	Routing
Basic Router Configuration Commands	
enable	Enter privileged EXEC mode
configure terminal	Enter global configuration mode
hostname <name>	Give the router a name
hostname Router1	
enable secret <password>	Set a password on the enable command
enable secret letmein	
interface <type><#>	Enter interface configuration mode
interface g0/1	
[no] ip address <address> <subnet mask>	Give an interface an IP address
ip address 10.1.1.1 255.255.255.0	Remove the current IP address
no ip address 10.1.1.1 255.255.255.0	
ip address dhcp	Set the interface to obtain an IP address from DHCP
no shut	Turn on the interface
end	Jump to the top of privileged EXEC
copy running-config startup-config	Save the running configuration
Router Serial Link Commands	
interface serial <#>	Enter serial interface configuration mode
interface s0/3/0	
encapsulation <hdlc ppp>	Default encapsulation is Cisco HDLC, choose PPP for vendor-neutral implementations
encapsulation ppp	
ip address 172.16.0.1 255.255.255.252	Configure interface IP address
clock rate <choose from list>	On DCE only: If desired, changelink speed from default 2,000,000 bps (range 300 – 4,000,000 bps)
no shut	Turn on the interface
show controller <controller number>	Shows if you are DCE or DTE on that link
show controller s0/3/0	Controller number = interface number
show interfaces <interface number>	Shows usual interface information including Layer 2 encapsulation protocol
show interfaces s0/3/0	
show ip interfaces brief	Display a list of all interfaces, their IP address, and their up/down status

Static Route Commands	
ip route <destination network> <destination subnet mask> <next hop outbound interface> [metric]	Add a static route For IPv6 – write mask in CIDR
ip route 172.16.3.0 255.255.255.0 192.168.2.254	IPv4 static route example
ipv6 route 2001:0db8:abcd::/64 2001:0db8:abcd:0000:0000:8a2e:0370:7334	IPv6 static route example Notice the subnet mask is in CIDR notation
ip route 172.16.3.0 255.255.255.0 192.168.2.254 200	IPv4 and IPv6 floating static route examples
ipv6 route 2001:0db8:abcd::/64 2001:0db8:abcd::8a2e:0370:7334 200	Metric (200) is set higher than AD of routing protocols
ip route 192.168.1.10 255.255.255.255 192.168.2.1	IPv4 host route example
ipv6 route 2601:140:8500:d3d0::7723/128 2001:0db8:abcd::8a2e:0370:7334	IPv6 host route example
ip route 0.0.0.0 0.0.0.0 < ip-address exit-intf >	Add an IPv4 default route
R1(config)# ip route 0.0.0.0 0.0.0.0 10.108.99.2	Default route to next hop
R1(config)# ip route 0.0.0.0 0.0.0.0 Serial0/1	Default route out serial interface on point-to-point WAN link
R1(config)# ipv6 route ::/0 2001::DB8:2:124 R1(config)# ipv6 route ::/0 Serial2/0	IPv6 default route examples
Router Show Commands	
show ip route	Display the IPv4 route table
show ip route ?	Find subcommands to help focus the route table output to subset of information
show ipv6 route	Display IPv6 route table entries
show ipv6 static	Display IPv6 static routes
show ip interfaces brief	List all interfaces, their status, and IPv4 address
show ipv6 interfaces brief	List all interfaces, their status, and IPv6 address
show running-config	Display the current running configuration
show startup-config	Display the current running configuration
show version	Display information about the router including hardware, IOS, and configuration register
Route Verification Commands	
show ip route	Display the IPv4 route table
show ipv6 route	Display the IPv6 route table
show ipv6 static	Display IPv6 static routes
ping <destination IPv4 address> ping 192.168.7.20	Prove that traffic can find a path to a destination and back
ping ipv6 <destination IPv6 address> ping 2001::DB8:2:124	
tracert ip <IPv4 destination address or hostname> tracert ip 192.168.7.20	Identify path to IPv4 destination node
tracert ipv6 <IPv6 destination address or hostname> tracert ipv6 2001::DB8:2:124	Identify path to IPv6 destination node

Debug Commands	
debug ip ?	List debug ip subcommands
debug ip [eigrp icmp nat ospf packet rip routing]	Debug a specific IP activity
no debug ip [eigrp icmp nat ospf packet rip routing]	Stop a running debug session
no debug all undebug all u all	Stop all running debug sessions
Routing Protocol Commands	
show running-config	Display the current running configuration
show ip route	Display the routing table
show ip protocols	Display information about any configured routing protocols
show ip rip	Show information about any RIP instance
show ip ospf	Show information about any OSPF instance
show ip eigrp	Show information about any EIGRP instance
show ip bgp	Show information about any BGP instance
Routing Table and Related Commands	
ip route 10.0.0.0 255.255.255.0 192.168.1.1 150	Configure a static route with a custom administrative distance
show ip route	Display the routing table
show ip route <destination> show ip route 173.45.67.0	Show routes for a specific network or prefix
show ip route <next-hop-ip> show ip route 10.1.1.254	Display routes learned from a specific next-hop IP address
show ip route interface <interface> show ip route interface g0/0	Display routes for a specific interface
show ip route summary	Display a summary of the routing table
ping <destination> ping 192.168.1.25	Send ICMP echo to a destination to proves network connectivity at Layer 3
ping	Extended ping; follow prompts
show ip interfaces brief	Show list of interfaces with status and IP
OSPF Commands	
interface Loopback 1 ip address 1.1.1.1 255.255.255.255 no shut	Create a loopback interface example
router ospf <process ID> router ospf 1	Enable OSPF and set the process ID
router-id <live IP address used to identify the router> router-id 1.1.1.1	Manually choose which interface IP to be the Router ID
network <connected network> <wildcard mask> area <#> network 1.1.1.1 0.0.0.0 area 0 network 77.88.0.0 0.0.255.255 area 0 network 66.55.44.0 0.0.0.255 area 1	Create a network statement

OSPF Show Commands	
show ip protocols	Display routing protocol configurations
show ip ospf	Display OSPF configurations and statistics
show ip ospf <process ID>	Display configs and statistics that OSPF process
show ip ospf 1	
show ip ospf interface	Display information on interfaces participating in OSPF: timers, role, and neighbor adjacencies
show ip ospf neighbor	Display OSPF neighbors, their ID, role, status
show ip route ospf	Display any routes in the learned by OSPF
show ip ospf border-routers	Display ABR and ASBR information
show ip ospf database	Display a summary of the OSPF database
OSPF Debugging Commands	
debug ip ospf adj	Show ospf neighbor adjacency events in realtime
debug ip ospf events	Show all ospf events as they happen
undebug all	Turn off debugging
NAT Commands	
ip nat inside	Specifies this interface is on the inside
ip nat outside	Specifies this interface is on the outside
access-list < 1 – 99 > permit < source IP> < wildcard mask >	Create a standard ACL allowing specified local
access-list 1 permit 192.168.1.0 0.0.0.255	addresses to be translated to global
ip nat pool < name > < start_address end_address > {netmask <subnet mask> prefix-length < bits >}	Create a pool of global addresses that can be used for translation
Router(config)# ip nat pool MY_POOL 203.0.113.2 203.0.113.62 netmask 255.255.255.192	
Router(config)# ip nat pool MY_POOL 203.0.113.2 203.0.113.62 prefix-length 26	
ip nat inside source list < ID > [interface pool] < name > overload	Enable PAT using an access list to permit certain inside clients to be translated
ip nat inside source list 1 interface g0/0 overload	PAT using one global address
ip nat inside source list 1 pool my_pool overload	PAT using global addresses from a pool
ip nat inside source static <inside IP> <outside IP>	Create a one-to-one static mapping
ip nat inside source static 192.168.1.10 203.0.113.10	
ip nat outside source list <ID> pool <name>	Enable outside-in NAT
ip nat outside source list 2 pool my_local_address_pool	
Verifying NAT	
show ip nat translations	Display the NAT table
show ip nat statistics	Display usage statistics for the NAT interfaces, access list and pool
clear ip nat translations	Clear the NAT table
debug ip nat	Debug NAT events
no debug ip nat	Stop debugging NAT events
undebug ip nat	Stop debugging NAT events
undebug all	Stop all debugging
u all	